Measuring Up

AP Human Geography · Unit 7 · One stimulus (like Question 2) · 7 points · Teacher key · GeoArcade FRQ Tutor

Geographers use many indicators to measure development, because no single number tells the whole story. The table shows six indicators for four countries.

Selected Development Indicators for Four Countries, 2023

Country	GNI per capita (PPP, US$)	HDI (0–1)	Gender Inequality Index (0–1, lower is better)	Workers in agriculture (%)	Workers in services (%)	Exports from top product (% of export value)
Ardena	48,200	0.93	0.08	2	78	9 (machinery)
Kovar	14,100	0.76	0.35	18	55	20 (electronics)
Sulmani	2,100	0.49	0.58	62	25	15 (coffee)
Tessar	31,400	0.74	0.50	6	60	85 (petroleum)

Note: The countries and data are invented for practice. Remember the class directions: label each part, and write Describe and Explain parts as T1- (your reason in general words), E2- (the specific proof), A3- (why it matters).

A. Identify the country in the table that is most likely in Rostow's stage of high mass consumption. IDENTIFY

EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:

- Ardena

MODEL ANSWER

Ardena.

Common mistake: "Tessar, because it has a high GNI per capita." GNI alone is the trap in this table. Tessar's HDI (0.74) is below Kovar's, even though Kovar earns less than half as much, and 85% of its exports are one raw material, so it isn't a mature, diversified economy. Check HDI and the services column too.

Why the model works: It names the one country whose numbers all point to the top stage: highest income, highest HDI, almost no farmers, and most workers in services.

B. Using the table, describe the relationship between the percent of workers in agriculture and the Human Development Index (HDI). DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- States the direction: higher percent in agriculture goes with lower HDI (inverse/negative relationship)
- Uses the table as proof: names countries or quotes numbers (e.g., Sulmani 62% / 0.49 vs. Ardena 2% / 0.93)

MODEL ANSWER

T1- The table shows a generally inverse relationship: the higher the percent of workers in agriculture, the lower the HDI tends to be.

E2- Sulmani has 62% of its workers in agriculture and the lowest HDI (0.49), while Ardena has only 2% in agriculture and the highest HDI (0.93).

A3- This fits the idea that as countries develop, workers shift out of farming into factories and services, which raises incomes, education, and life expectancy.

Common mistake: "Agriculture and HDI are related." Related how? A relationship has a direction (as one goes up, the other goes down), and you need numbers from the table as proof.

Why the model works: It states the direction (inverse), proves it with two rows and their numbers, and explains why farming and HDI move in opposite directions.

C. Using the table, explain why a country can have a high GNI per capita but a lower HDI than its income would predict. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a reason: wealth is concentrated or unequal, money comes from one resource (oil), low spending or corruption, or gender inequality
- Explains the link to HDI: money doesn't reach education, health, or life expectancy for most people

MODEL ANSWER

T1- A country can have a high GNI per capita but a lower HDI when its wealth is concentrated in a few hands and not spent on people.

E2- Tessar earns $31,400 per person on paper, but 85% of its exports are petroleum, and oil money often flows to the government and a small elite rather than into schools and hospitals; its high Gender Inequality Index (0.50) suggests that women in particular are left out.

A3- Because HDI also measures education and life expectancy, not just income, a country that does not invest its oil money in health and schooling scores lower than its GNI alone would predict.

Common mistake: “Tessar has a lower HDI because it is a smaller country.” Size has nothing to do with it; the table doesn't even show population. The reason is WHERE the money goes: concentrated oil wealth that isn't spent on schools and health.

Why the model works: It names the reason (concentrated wealth), proves it with Tessar's row (85% petroleum, GII 0.50), and connects it to what HDI measures (education and life expectancy).

D. Explain ONE way a high Gender Inequality Index can slow a country's economic development. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a specific barrier women face: schooling, paid jobs, loans/credit, property, political power, health, early marriage or childbirth
- Explains the effect on the economy: fewer workers or skills, less produced, lower incomes, slower growth

MODEL ANSWER

T1- High gender inequality (a high GII) can slow development because it keeps women, about half the population, from fully contributing to the economy.
E2- A GII as high as Sulmani's (0.58) signals that fewer girls finish secondary school, more have children as teenagers, and fewer women are in the labor force.
A3- Because educated, working women earn income, have fewer and healthier children, and start businesses, shutting them out shrinks the workforce and the country's skills, so the economy grows more slowly than it could.

Common mistake: “Gender inequality is bad for the economy.” That just restates the question. Name a specific thing women are kept from (school, jobs, loans) and explain how that leads to slower growth.

Why the model works: It names the general idea (half the population held back), gives specific barriers that Sulmani's high GII signals (school, teen births, work), and explains the chain to slower growth.

E. Explain ONE economic risk for a country whose exports depend heavily on a single commodity. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a risk: price shock or volatility, running out of the resource, falling demand, or the resource curse
- Explains the effect on the country: lost income or revenue, budget cuts, debt, unemployment, recession

MODEL ANSWER

T1- Depending on one commodity is risky because the whole economy rises and falls with that product's world price, which the country cannot control.
E2- Tessar gets 85% of its export earnings from petroleum, so if oil prices crash, as they did from over $100 a barrel in 2014 to under $30 in 2016, its export earnings, government revenue, and oil jobs can all fall sharply together.
A3- Because the country has few other exports to fall back on, a price shock can lead to budget cuts, debt, and unemployment, which is one part of what geographers call the resource curse: resource-rich countries often end up less stable and less developed than expected.

Common mistake: “Tessar should sell more things.” That's advice, not a risk. The question asks what could go WRONG: name the danger (a price crash) and explain how it hurts the country.

Why the model works: It names the risk (price shock), gives proof (Tessar's 85% petroleum and the 2014–2016 oil crash), and traces the chain to budget cuts, debt, and job losses.

F. Explain how a special economic zone (SEZ) or export processing zone could help Sulmani move toward Rostow's take-off stage, AND identify one potential downside. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Explains what the SEZ offers or attracts: tax breaks, no tariffs, foreign investment, factories, jobs, exports, technology
- Connects it to take-off or industrial growth: moving workers out of farming, a new leading industry, faster growth
- Identifies a downside: low wages, exploitation, pollution, weak protections, lost tax revenue, companies leaving, dependence on foreign firms

MODEL ANSWER

T1- A special economic zone could help Sulmani start to industrialize by attracting foreign factories.
E2- Inside the zone, the government offers low taxes, no tariffs on imported parts, and ready-built roads and power, so multinational companies open clothing or electronics plants that hire workers leaving farms, much as the Shenzhen SEZ turned a small border town into a manufacturing giant.
A3- Those factories create a new leading sector, export earnings, and job skills that can push Sulmani from Rostow's traditional stage toward take-off; however, the downside is that wages are often low and labor and environmental rules are weak, and if the tax breaks end the companies may simply move to a cheaper country.

Common mistake: "An SEZ would give Sulmani factories and jobs." That's a start, but it stops too early. Connect the factories to take-off (workers leave farms, industry grows fast) AND name one downside.

Why the model works: It explains what the zone offers (tax breaks, infrastructure, foreign factories), connects it to take-off (new leading sector pulling workers from farms), and names a clear downside (low wages, companies leaving).

G. Explain ONE limitation of using the data in the table to compare the quality of life of people in these four countries. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a limitation: national figures hide regional/urban-rural/gender/rich-poor gaps, informal economy undercounted, only one year, PPP issues, or quality of life factors not measured
- Explains why it misleads, ideally with an example from the table

MODEL ANSWER

T1- One limitation is that every number in the table is a single national figure for a single year, which hides differences between groups inside each country.
E2- For example, Tessar's $31,400 GNI per capita is an average: a small oil elite in the capital may be extremely rich while rural families and women (GII 0.50) live on far less, and Kovar's 55% service workers may be concentrated in a few big cities.
A3- Because national figures flatten the gaps between rich and poor regions, genders, and the formal and informal economy, two countries with similar numbers could offer very different lives, so you would need data at a smaller scale to compare quality of life fairly.

Common mistake: “The table only has four countries.” That's about how many rows there are, not about what the numbers can tell you. Explain what the national numbers hide, like gaps between rich and poor regions or men and women.

Why the model works: It names the limitation (national figures for one year), shows what they hide with a specific example (Tessar's oil elite vs. rural families), and explains why that makes comparing quality of life unreliable.

Scoring guide written for the GeoArcade FRQ Tutor. Award each part independently, 1 point each.